

Exercise Week 1

Today's exercises

Exercise 1-1: Blink LED on microcontroller board

Exercise 1-2: Blink LED on breadboard

Exercise 1-3: Display messages on serial monitor

Exercise 1-4: Variable resistor

Exercise 1-5: PWM signal

Exercise 1-6: Photointerrupter

Exercise 1-7: Photointerrupter & LED

Show the result of each exercise to an instructor or TA, every time you complete an exercise

Procedure

Step 1: Install **Visual Studio (VS) Code** on your PC

- See P. 2-3 of “**platformIO.pdf**” in “**manual**” folder on Teams
- Skip this step if it is already installed.

Step 2: Install **PlatformIO IDE** for VS Code

- See P. 5 of “**platformIO.pdf**”

Step 3: Make a **new project**

- See P. 7-10 of “**platformIO.pdf**”

Step 4: Write a **code** (see P. 4 of **this document**)

Step 5: Connect a microcontroller to your PC

Step 6: Build & run the project (see P. 14 of “**platformIO.pdf**”)

- See P. 16-18 of “**platformIO.pdf**”, if the upload fails

Example code

```
#include "mbed.h"

// Blinking rate in milliseconds
#define BLINKING_RATE 500ms

int main()
{
    DigitalOut led(LED1);

    while (true) {
        led = !led;
        ThisThread::sleep_for(BLINKING_RATE);
    }
}
```

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Exercise 1-7: Photointerrupter & LED

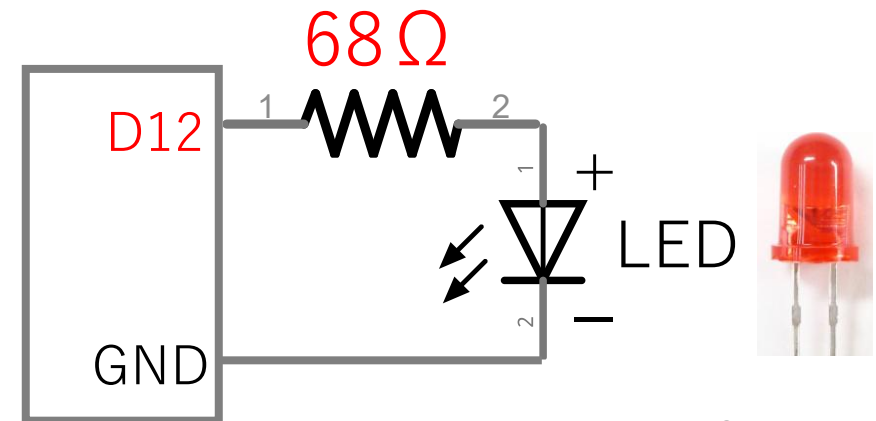
Show the result of each exercise to an instructor or TA, every time you complete an exercise

Exercise 1-2

Make a circuit in the figure on a breadboard and blink an LED (red) using **D12** pin of a microcontroller board. Then, show the result and **explain**

- why a resistor is necessary
 - how to select the resistance value ($68\ \Omega$)
 - what the forward voltage of an LED is
- to an instructor or a TA

- See the next page for the procedure
- Ask an instructor or TA questions, if you don't understand



Procedure

Step 1: See P. 8-10 to learn about the LED

Step 2: See P. 11-12 to learn about breadboards

Step 3: Make a circuit on your breadboard (P. 13-14)

Step 4: Change the **pin name** in main.cpp of Exercise 1-1 (P. 15)

Step 5: Connect a microcontroller to your PC

Step 6: Build & run the project (same as in Exercise 1-1)

- See P. 14 of “**platformIO.pdf**”

- See P. 16-18 of “**platformIO.pdf**”, if the upload fails

Blink a LED on breadboard



- Polarity (difference of “+” and “-”)
- Maximum current 30mA (from [data sheet](#))

Symbol

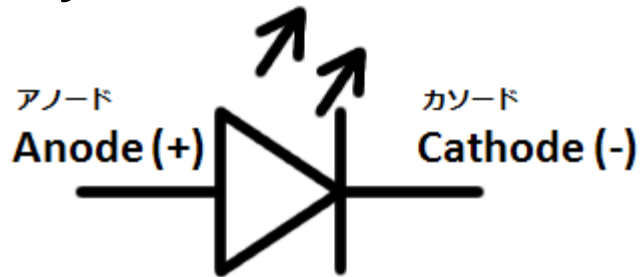
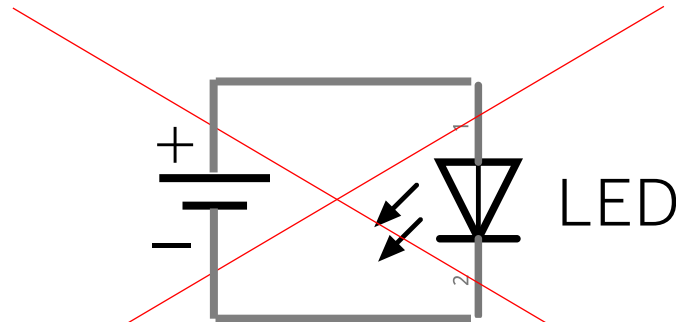
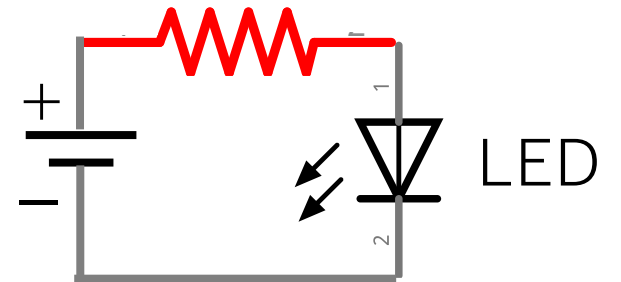
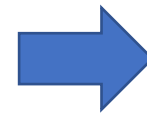


Image: <https://digikay.com/>



Too much current
("short" circuit)



Use a resistor to
reduce the current

Polarity: 極性, anode: アノード, cathode: カソード, resistor: 抵抗

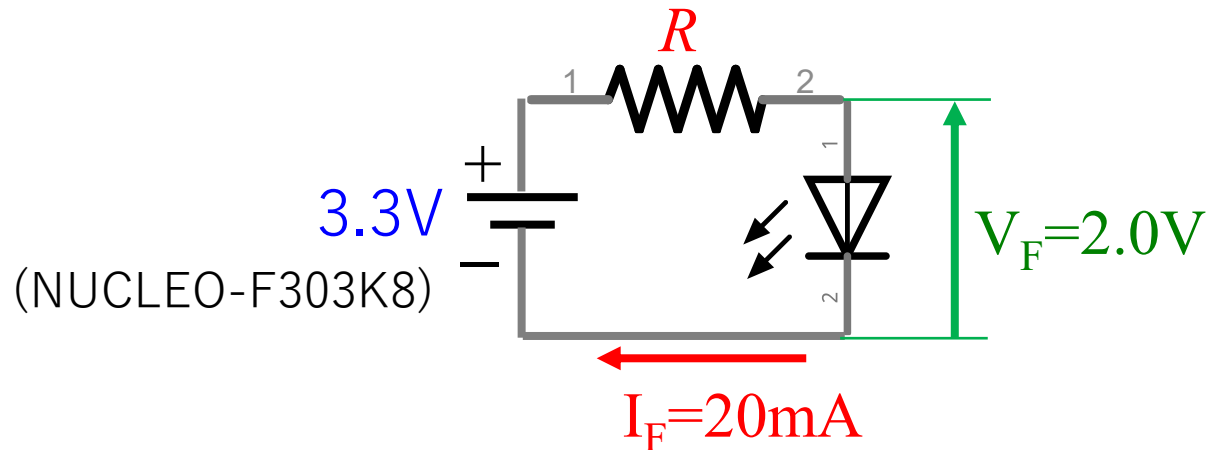
Calculating the necessary resistor value

■Electrical -Optical Characteristics

(Ta=25°C)

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
DC <u>Forward Voltage</u> *1	V _F	I _F =20mA	1.8	2.0	2.5	V

From the data sheet



From Ohm's law,

$$R = \frac{3.3 - 2.0}{20 \times 10^{-3}} = 65\Omega$$

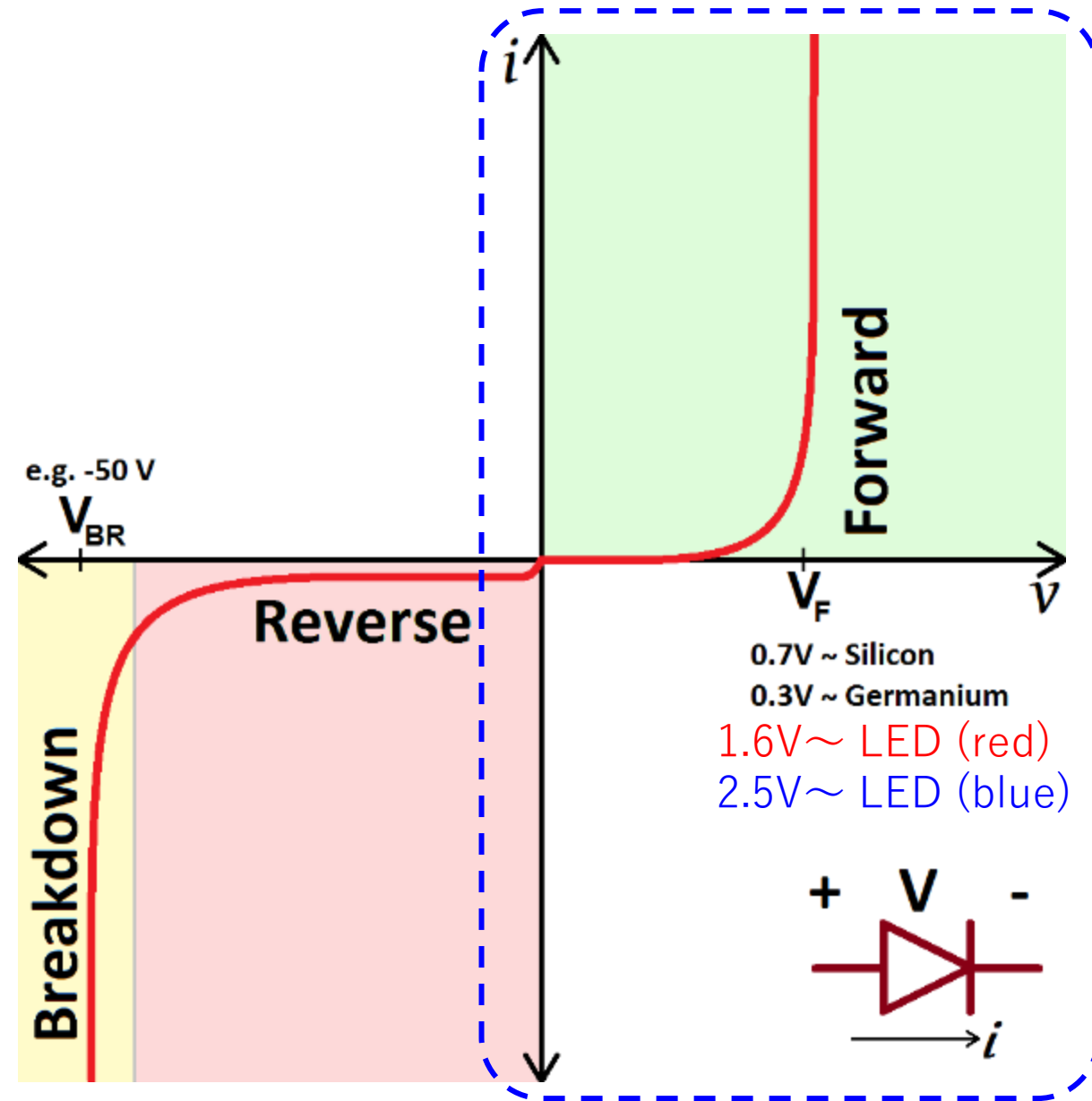
Let's use 68Ω

Available in
the market

Forward voltage: 順方向電圧

Forward voltage

- LED (Light-Emitting Diode) is a special type of diode
- If $V < V_F$ (forward voltage), almost no current flows
- If $V > V_F$, current rapidly increases as V
- The value of V_F depends on types of diodes



Light-emitting diode: 発光ダイオード

Typical current-voltage relationship of a diode.

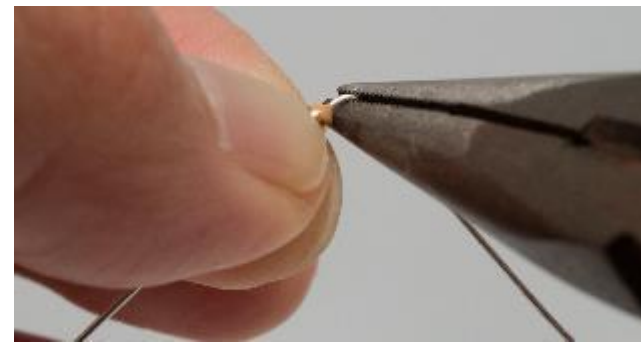
Image source: <https://learn.sparkfun.com/>

Breadboard

Connected

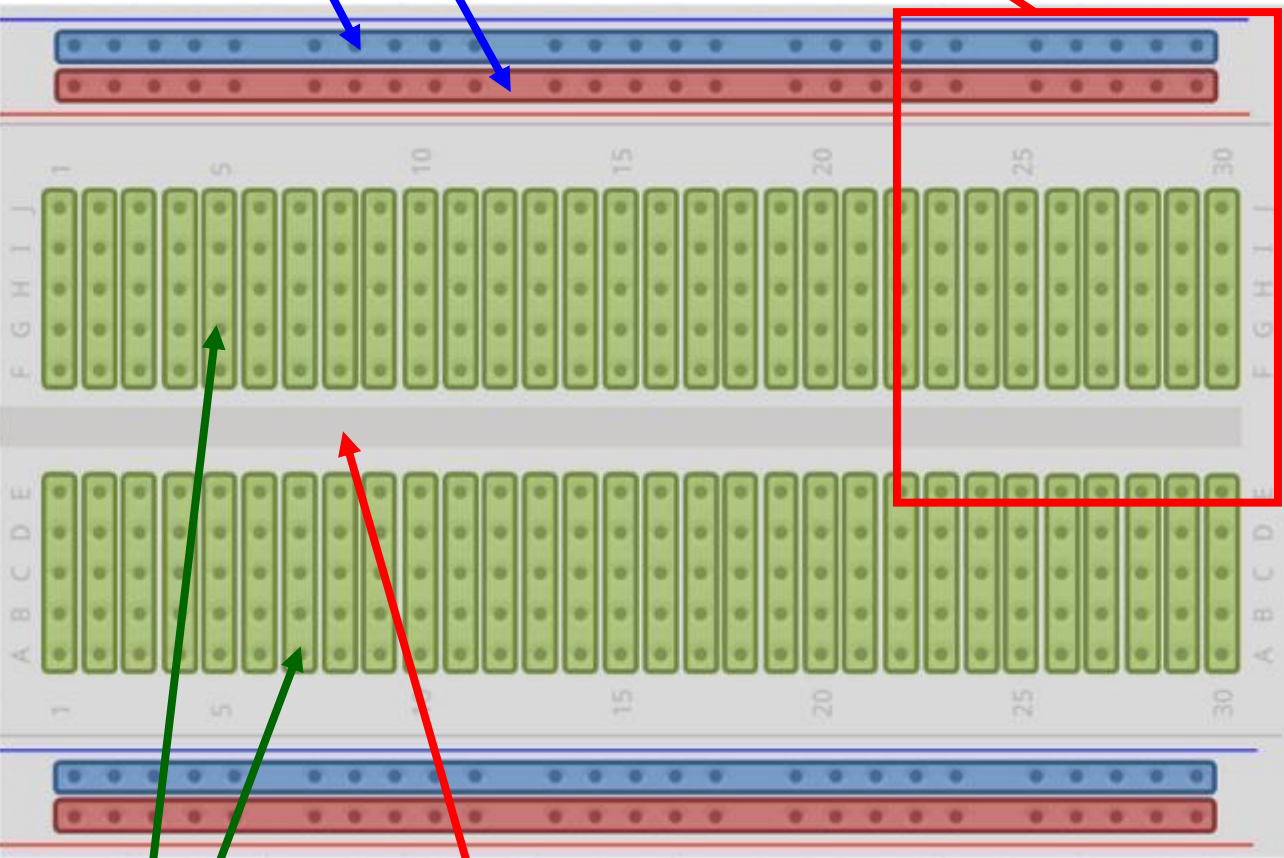


Back side



Bend lead wires of resistors to insert them into a breadboard

Image: <https://synapse.kyoto/>



Connected Not Connected

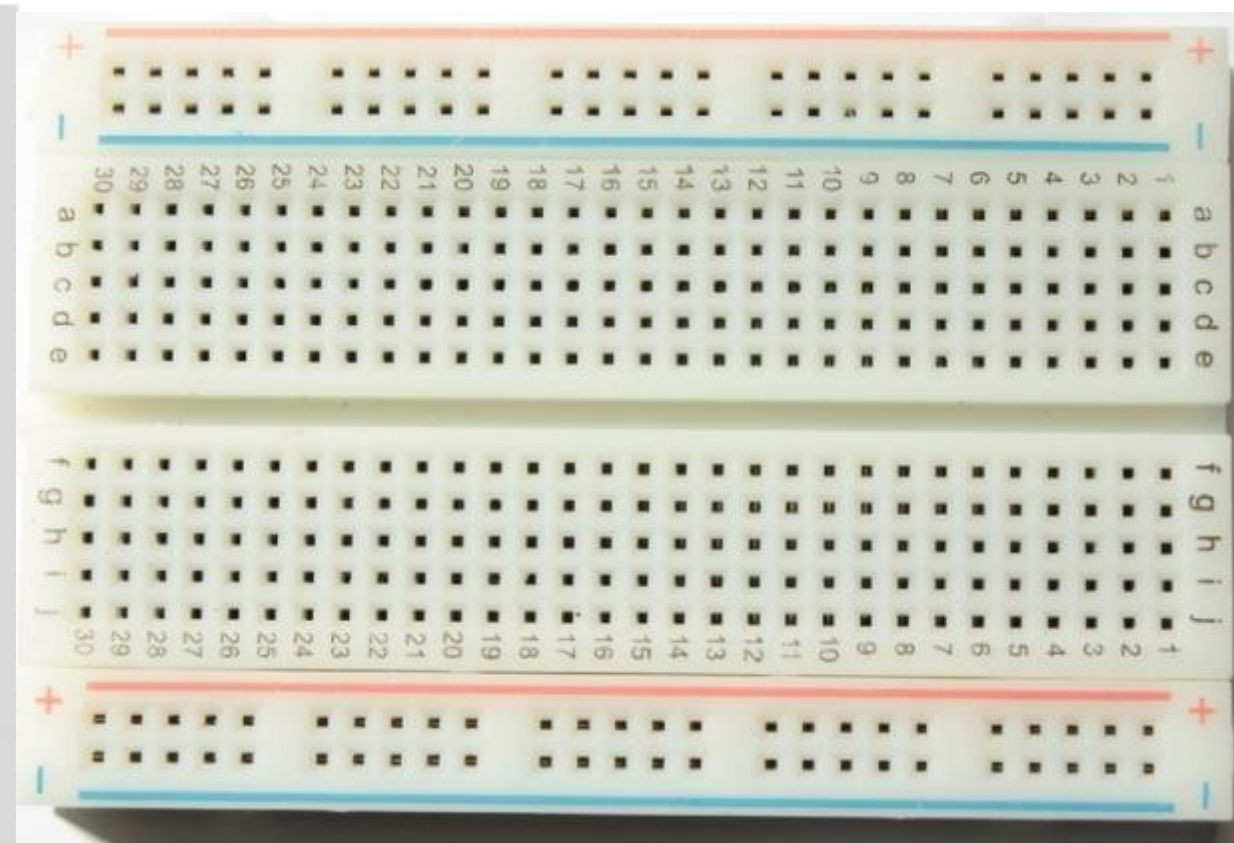
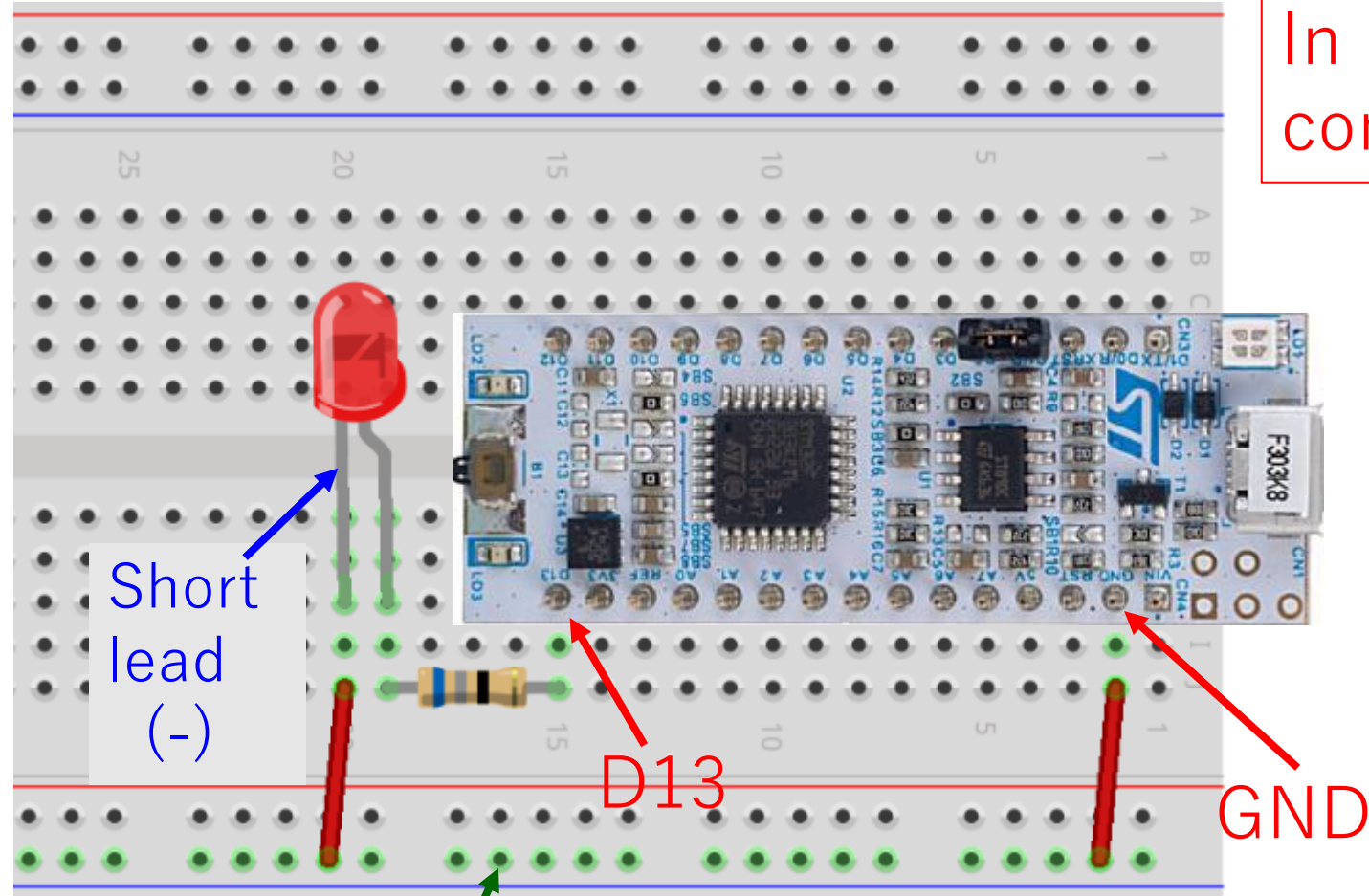


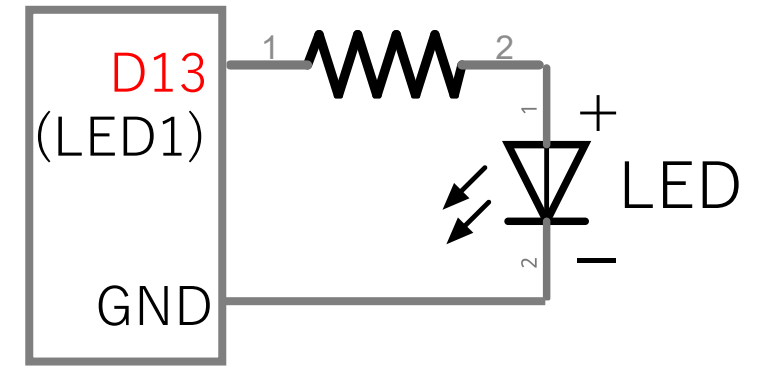
Image: <https://nobita-rx7.hatenablog.com/>

Example

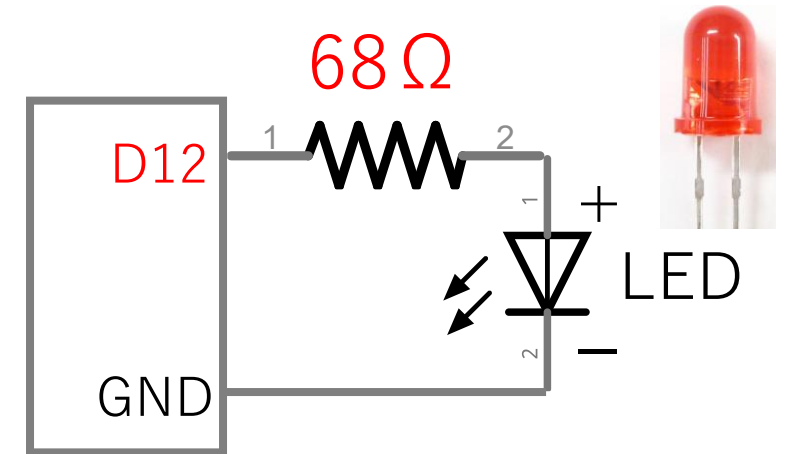
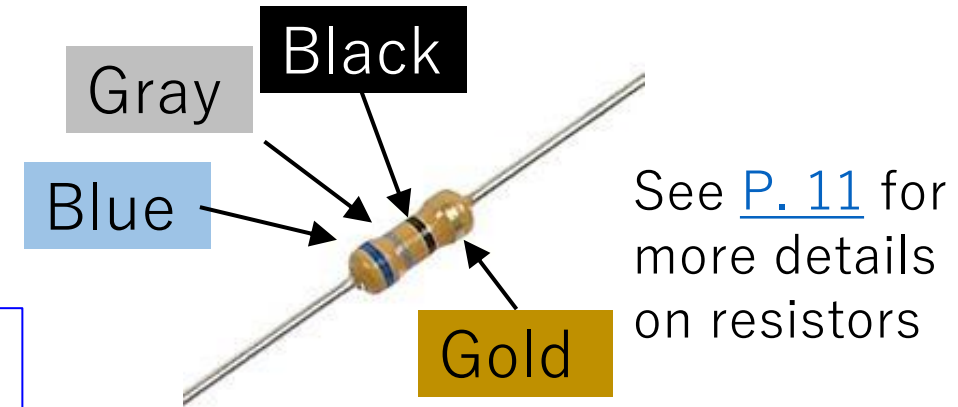
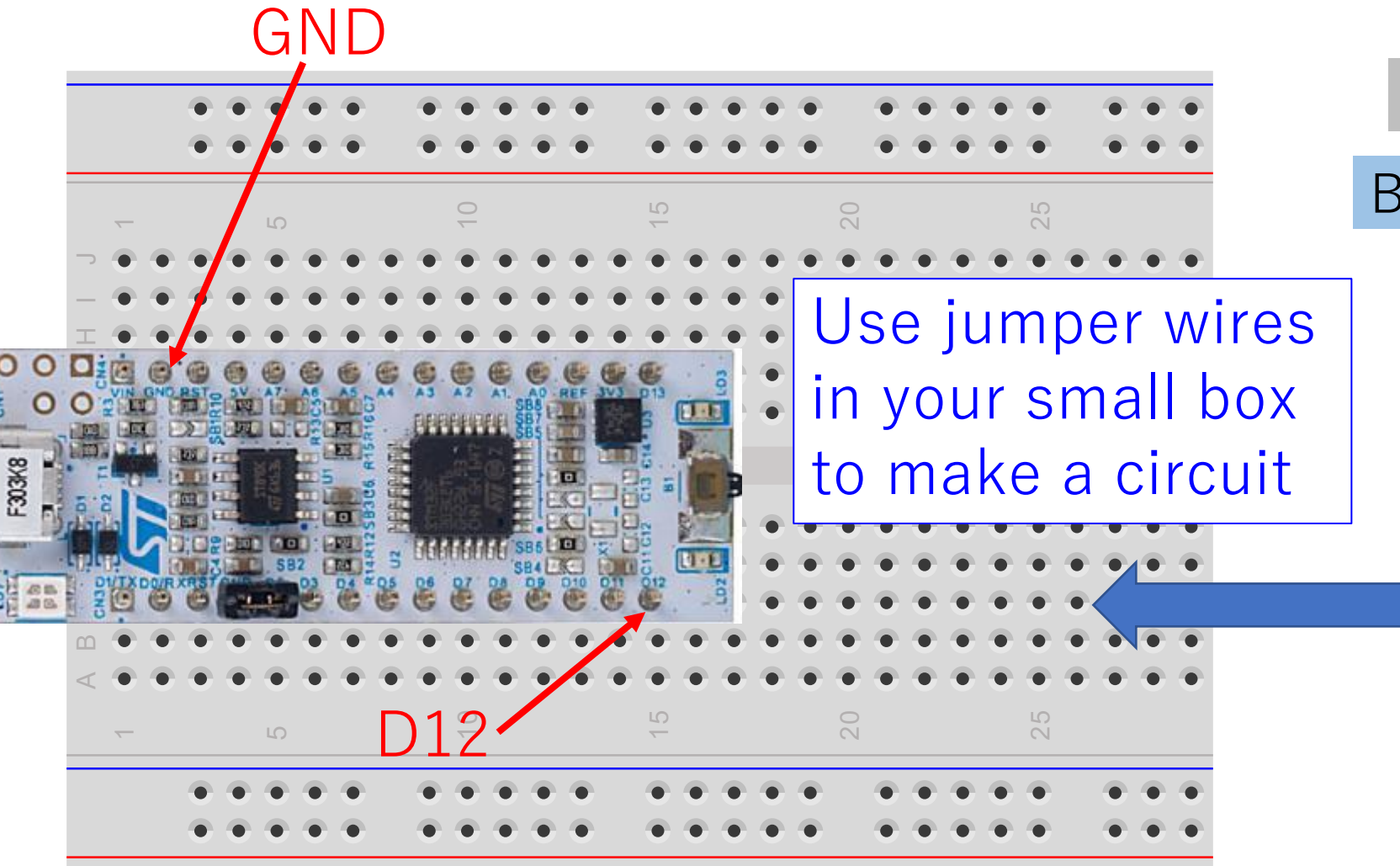
In Exercise 1-2, you need to connect **D12** instead of D13!!



Green dots illustrate electrically connected holes

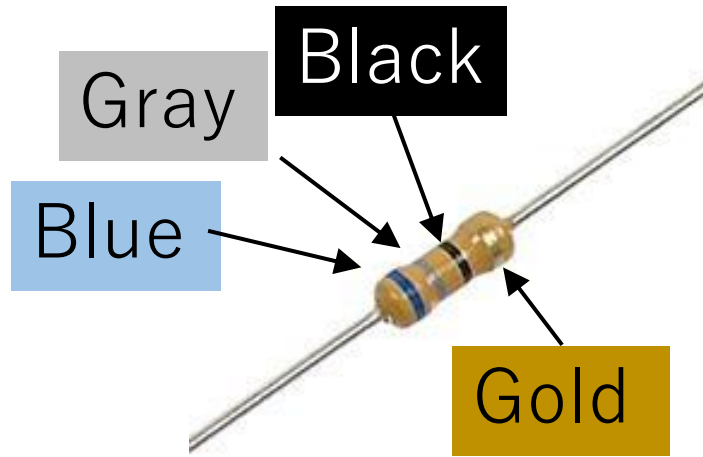


Step 2: Make a circuit on your breadboard



Resistors

Example: 68 Ω

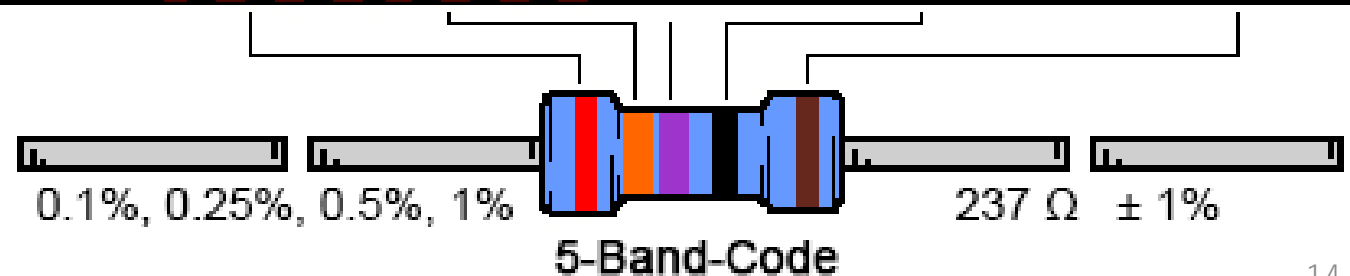


Blue Gray Black
 68×10^0

Tolerance: ± 5 (%)
許容差

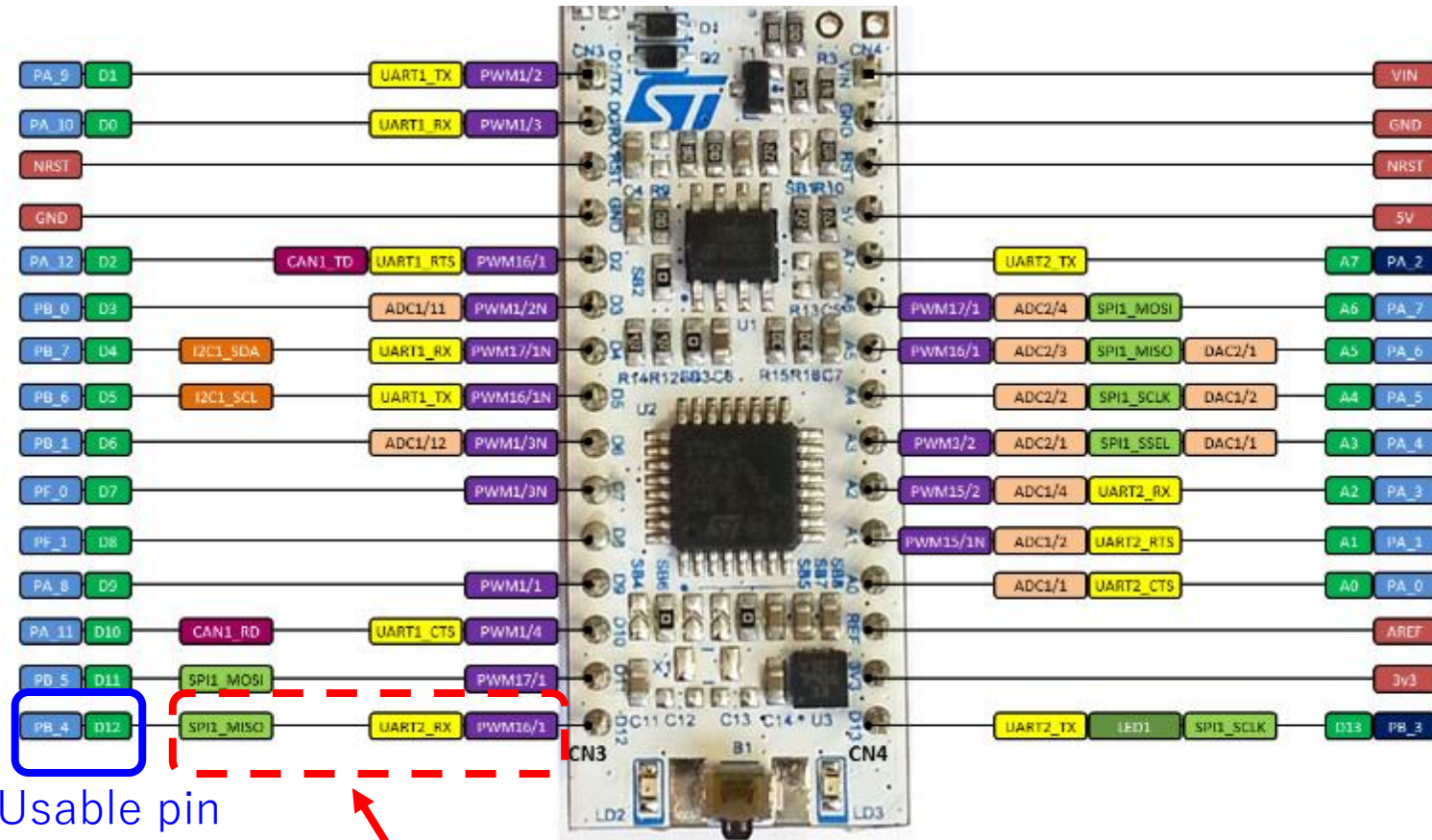


COLOR	1 ST BAND	2 ND BAND	3 RD BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1Ω	
Brown	1	1	1	10Ω	± 1% (F)
Red	2	2	2	100Ω	± 2% (G)
Orange	3	3	3	1KΩ	
Yellow	4	4	4	10KΩ	
Green	5	5	5	100KΩ	± 0.5% (D)
Blue	6	6	6	1MΩ	± 0.25% (C)
Violet	7	7	7	10MΩ	± 0.10% (B)
Grey	8	8	8	100MΩ	± 0.05%
White	9	9	9	1GΩ	
Gold				0.1Ω	± 5% (J)
Silver				0.01Ω	± 10% (K)



Step 3: Change the **pin name** in the previous program

You need to change to D12 or PB_4



Usable pin names are "D12" and "PB_4"

You cannot use these pin names in your program

```
13  int main()
14  {
15      // Initialise the digital pin
16      DigitalOut led(LED1);
17
18      while (true) {
19          led = !led;
20          ThisThread::sleep_1000;
21      }
22  }
```

main.cpp of "mbed-os- example-blinky"
See also P. 28 of "mechatro1_week1.pdf"

Today's exercises

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Exercise 1-2: Blink LED on breadboard

Exercise 1-3 : Display messages on serial monitor

Exercise 1-4: Variable resistor

Exercise 1-5: PWM signal

Exercise 1-6: Photointerrupter

Exercise 1-7: Photointerrupter & LED

Show the result of each exercise to an instructor or TA, every time you complete an exercise

Exercise 1-3

On a **terminal** of PlatformIO, display “LED ON” when LED is ON and “LED OFF” when LED is OFF. Then, show the result to an instructor or a TA

Procedure:

Step 1: Modify the previous program (P. 18)

Step 2: Build & run the project (same as in Exercise 1-1)

- See P. 14 of “**platformIO.pdf**”

- See P. 16-18 of “**platformIO.pdf**”, if upload fails

Step 3: Check serial output in the terminal (P. 19)

Step 1: Modify the previous program

```
11  int main() {  
12      // Initialise the digital pin D12 as an output  
13      DigitalOut led(D12);  
14  
15      while (true) {  
16          if (led == 0) {  
17              led = 1; // Turn LED on  
18              printf("LED ON\r\n");  
19          }  
20          else {  
21              led = 0; // Turn LED off  
22              printf("LED OFF\r\n");  
23          }  
24          ThisThread::sleep_for(BLINKING_RATE);  
25      }  
26  }
```

Modify this part
of the program
for Exercise 1-2



Step 3: Check the serial monitor

Displays serial output in a terminal

```
1  #include "mbed.h"
2
3  #define BLINKING_RATE 500ms
4
5  int main()
6  {
7      DigitalOut led(D12);
8
9      while (true){
10         if (led == 0){
11             led = 1;
12
13             printf("LED OFF\r\n");
14         }
15         ThisThread::sleep_for(BLINKING_RATE);
16     }
17 }
```

Terminal

Executing task in folder ex1-3: C:\Users\fuku2\.p1
atformio\penv\Scripts\platformio.exe device monitor

LED OFF
LED ON
LED OFF
LED ON
LED OFF
LED ON
LED OFF
LED ON
LED OFF
LED ON
LED OFF
LED ON

Today's exercises

Exercise 1-1: Blink LED on microcontroller board

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Exercise 1-5: PWM signal

Exercise 1-6: Photointerrupter

Exercise 1-7: Photointerrupter & LED

Show the result of each exercise to an instructor or TA, every time you complete an exercise

Exercise 1-4

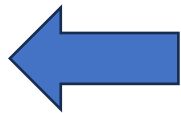
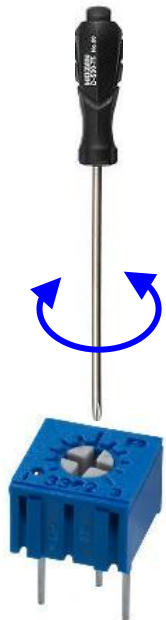
- Add a **variable resistor** (see P.22) to the circuit in Exercise 1-3
- Change the resistance by turning the screw
- Show the result to an instructor or a TA

3362P-1-104LF

Mark "P 104"
on a surface

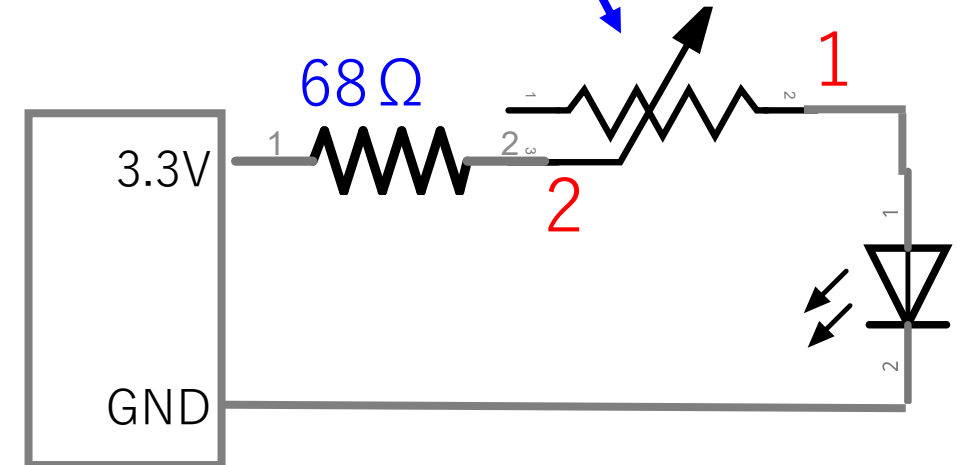


$R = 0 \sim 100 \text{ k}\Omega$

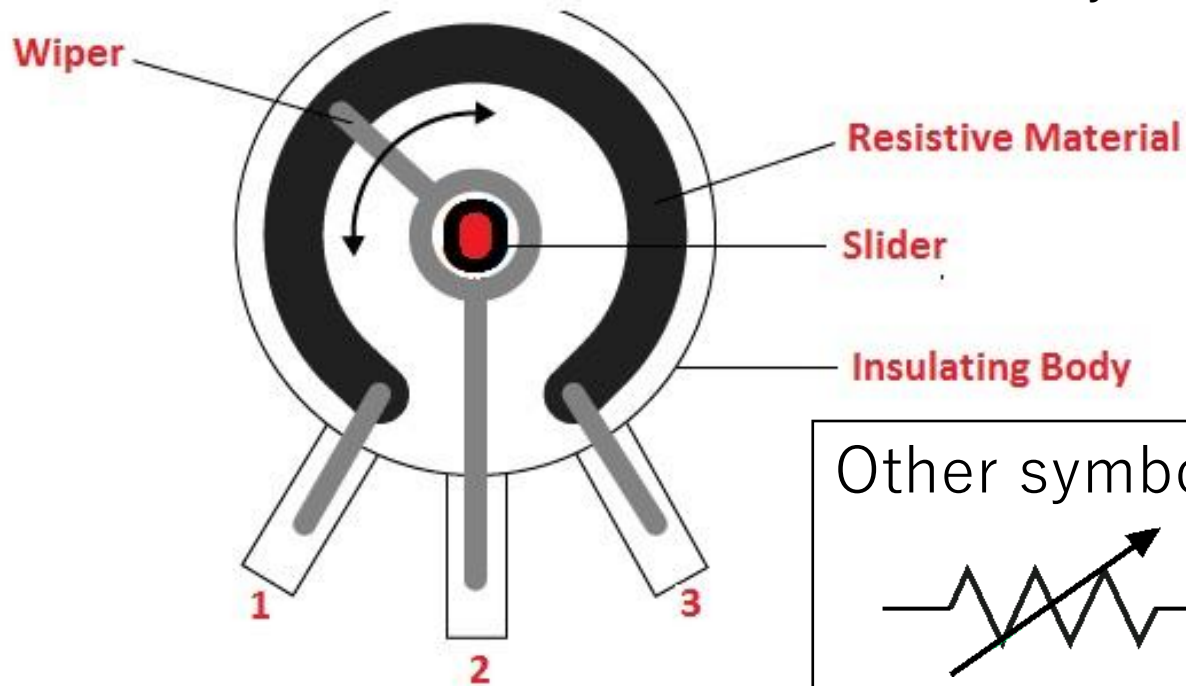
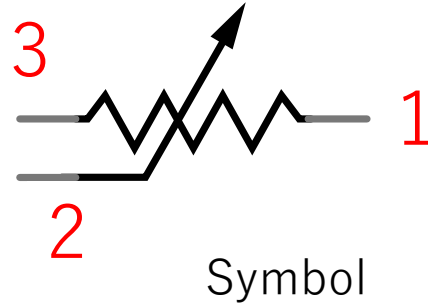


Tool set

Get one from a TA
and **return at the end of the class**



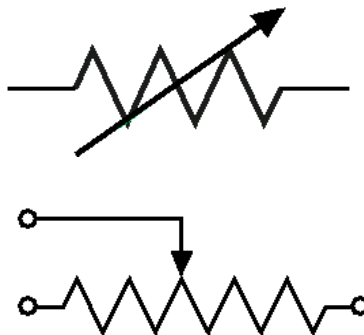
Variable resistors



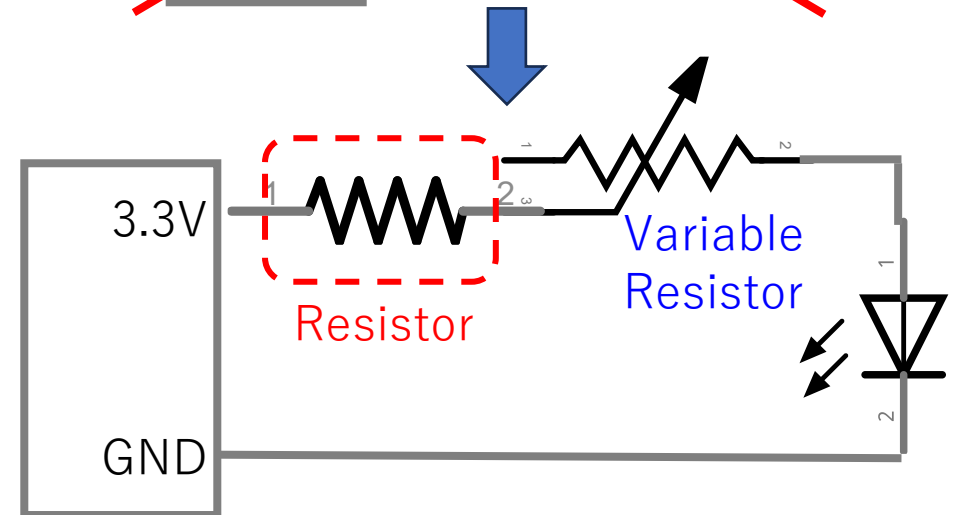
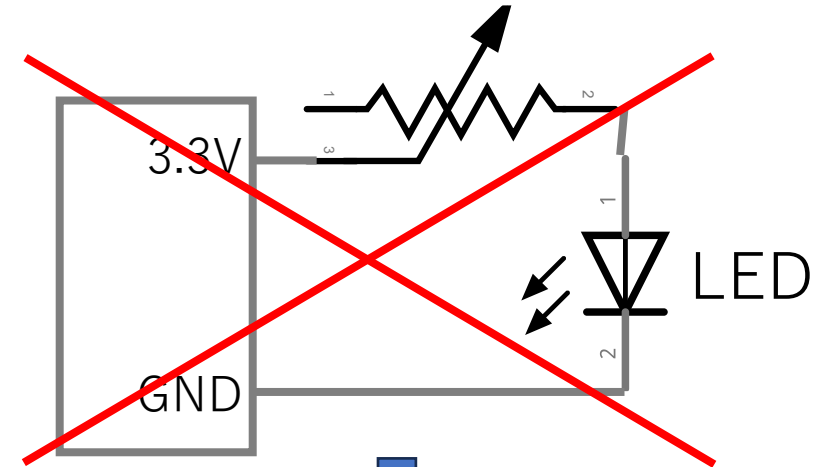
Terminals in a Variable Resistor

www.CircuitsToday.com

Other symbols



Use it with another resistor to avoid too a large current



Variable resistor: 可變抵抗, terminal: 端子

Today's exercises

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Exercise 1-4: Variable resistor

Exercise 1-5: PWM signal

Exercise 1-6: Photointerrupter

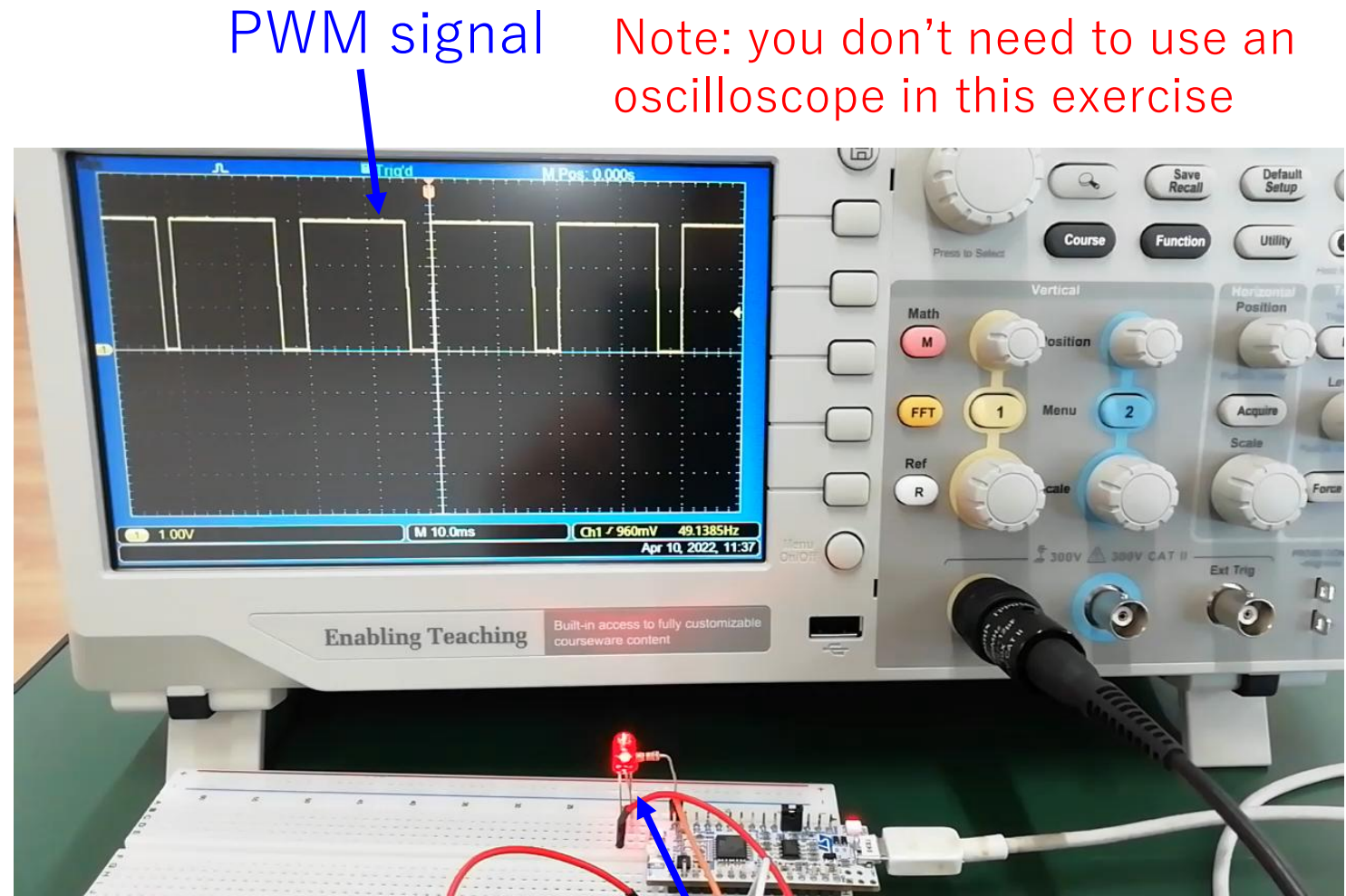
Exercise 1-7: Photointerrupter & LED

Show the result of each exercise to an instructor or TA, every time you complete an exercise

Exercise 1-5

- Repeat turning a LED ON and OFF with the period 2.0 sec by **gradually changing** the **PWM signals** from a microcontroller.
- Show the result and **explain your code to an instructor or a TA**

➤ See NEXT page for the procedure



Note: you don't need to use an oscilloscope in this exercise

[Link to the movie](#)

LED gradually changing the brightness

Procedure

Step 1: See NEXT page to learn about PWM signals

Step 2: Choose a pin of microcontroller used for PWM (P. 27)

Step 3: Make a new project (same as in Exercise 1-1)

➤ See P. 7-10 of “platformIO.pdf”

Step 4: Write a code in main.cpp

➤ Use an Mbed API “PWMOut” (P. 28) to generate PWM signals

➤ Gradually increase the “duty cycle” of PWM signals

Step 5: Build & run the project (same as in Exercise 1-1)

Example:

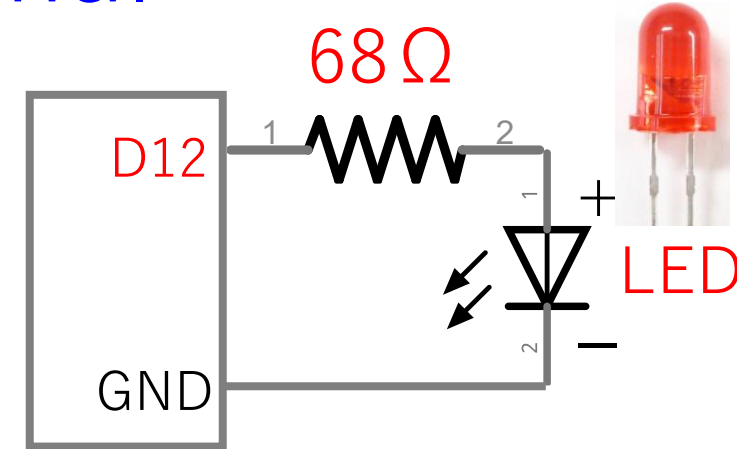
- Increase duty cycle by 10% every 0.1 sec, using a “for-loop”.
- Then decrease duty cycle by 10% every 0.1 sec using another “for-loop” (P. 29).

PWM (Pulse-width modulation) signal

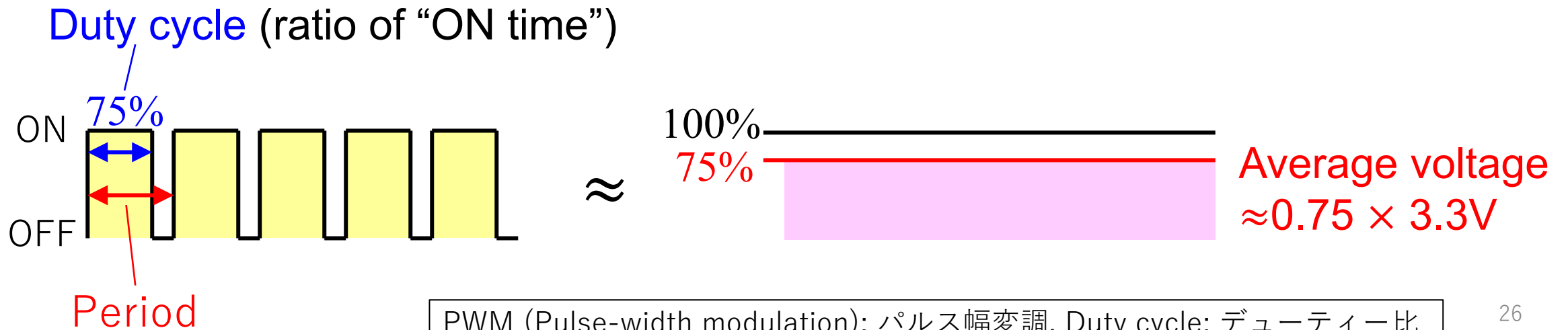
Voltage from a microcontroller pin is fixed ($\approx 3.3\text{V}$)
→ How do you change the brightness of a LED ?



Switch ON&OFF voltage supply for a **short period**
→ **Duty Cycle** $\times 3.3\text{V} \approx \text{Average Voltage}$

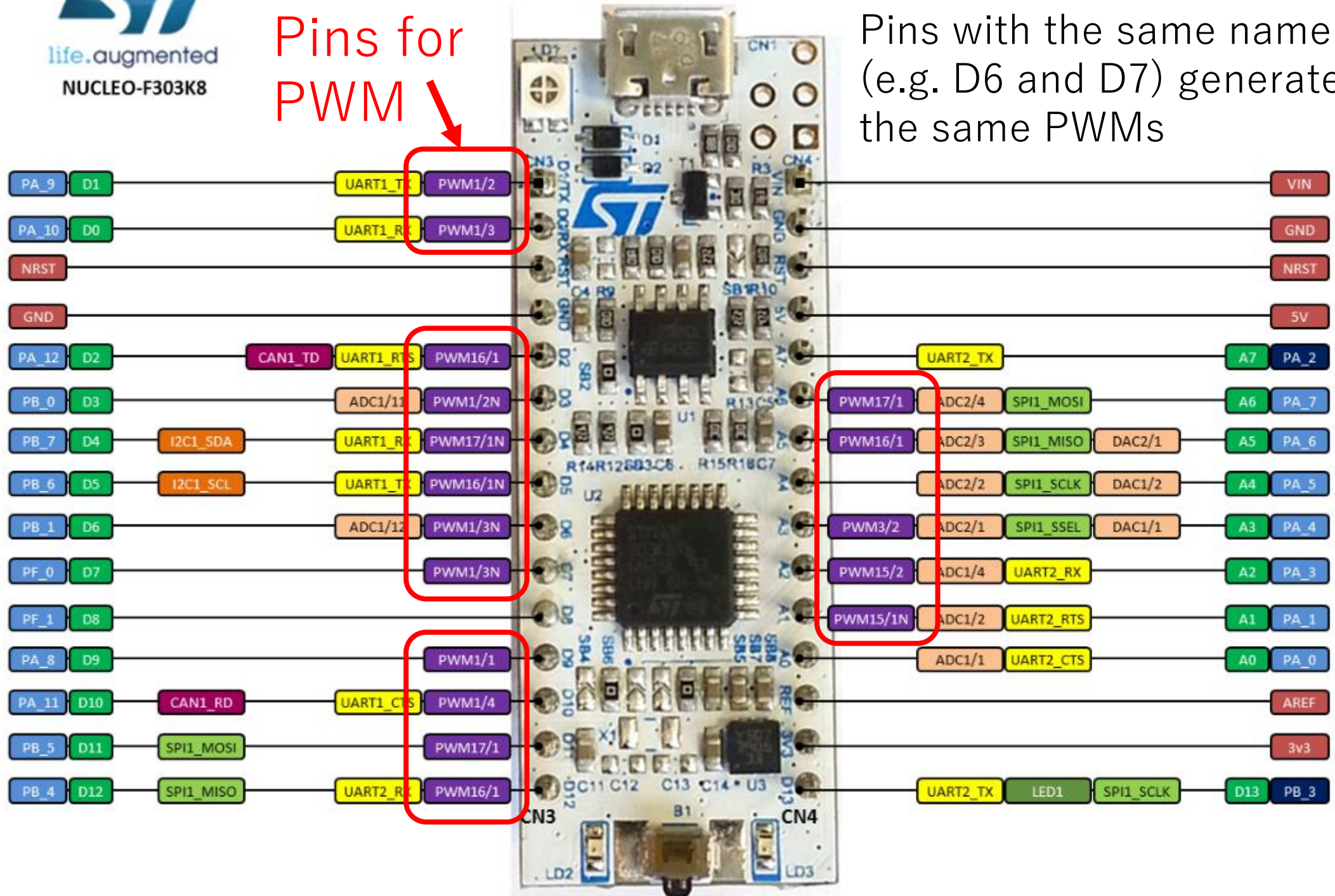


Example of PWM signal



Pins for
PWM

Pins with the same name
(e.g. D6 and D7) generate
the same PWMs



Mbed API: “PWMOut”

[Click for Documentation](#)

Initialization:

```
PWMOut object_name(Pin_name);
```

Class

Functions of PWMOut :

- period()
- write()
- read(), etc



Example:

Set the **PWM period** as 0.02 sec
And the **duty cycle** as 3%

```
object_name.period(0.02f)  
object_name.write(0.03f)
```

- The default period is 0.020s, and the default duty cycle is 0.
- “0.02f” means 0.02 of the float type.

For-loop

Syntax:

```
for (Initialization; condition; update) {  
    // code to be executed  
}
```

- Example 1 (integer):

```
for (int i = 0; i < 10; i++) {  
    // code to be executed  
}
```

Useful for
Exercise 1-5



- Example 2 (float):

```
for (float p = 0.0f; p <= 1.0f; p+=0.1f) {  
    // code to be executed  
}
```

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Show the result of each exercise to an instructor or TA, every time you complete an exercise

Exercise 1-5

Make a circuit and a program that indicates

- “black” if a black line is in front of a **photointerrupter**
- “white” if not

on a terminal (P. 19). Then, show the result and **explain**

- **Purposes of the two resistors in the circuit**
- **What is “collector current” on P. 41**
- **Purpose of AnalogIN API**

to an instructor or a TA

➤ See P. 32-33 for the procedure

Procedure -- Part 1--

Step 1: See P. 34–41 to learn how to use the [photointerrupter](#) and “[AnalogIn](#)” API

Step 2: Choose a pin of microcontroller used for ADC (P. 38)

Step 3: Make a circuit on a breadboard (P. 42)

Step 4: Make a new project (same as in Exercise 1-1)

- See P. 7-10 of “[platformIO.pdf](#)”

Step 5: Setup to [display decimal numbers](#) in the terminal

- See P. 22 of “[platformIO.pdf](#)”

Step 6: Write a code in main.cpp

- Use “[AnalogIn](#)” API to read the voltage from the pin

- Use printf function to show the value from AnalogIn in the terminal (See P. 43 about how to use)

Procedure -- Part 2--

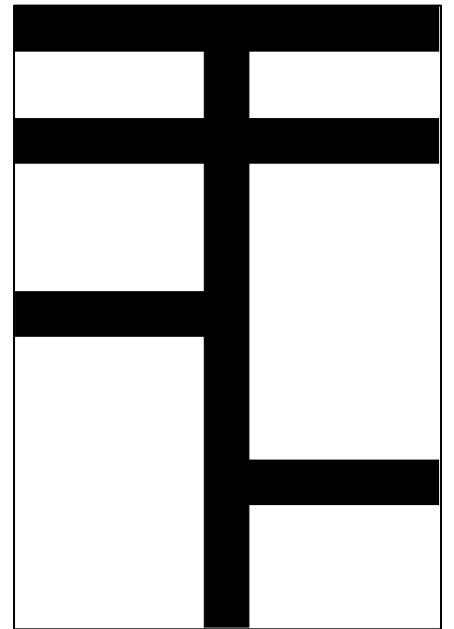
Step 7: Build & run the program, then observe the values on the terminal when a black or white surface is in front of the sensor. Use a sheet of paper with black lines in your box.

Step 8: Choose a threshold between “black” and “white”

Step 9: Add an IF-THEN statement to your program to indicate

- “black” if a black line is in front of a photointerrupter
- “white” if not

using the threshold chosen in Step 8

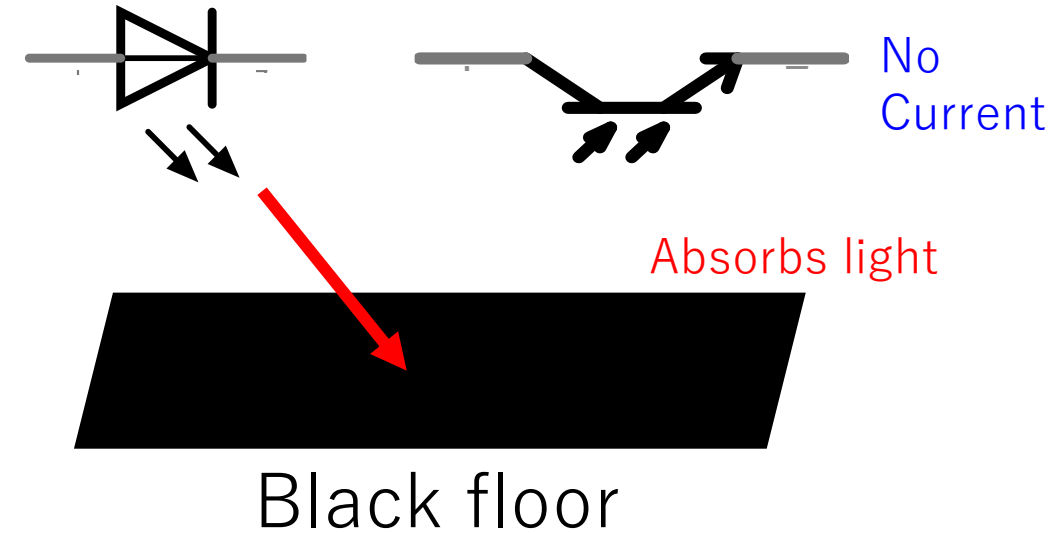
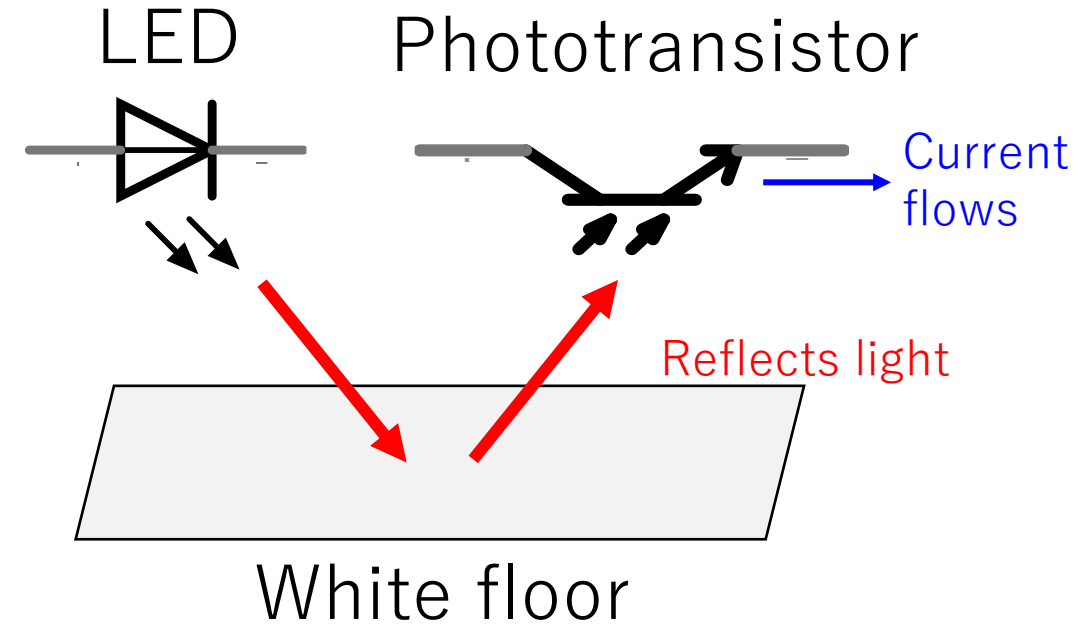


Photointerrupter (reflective type)



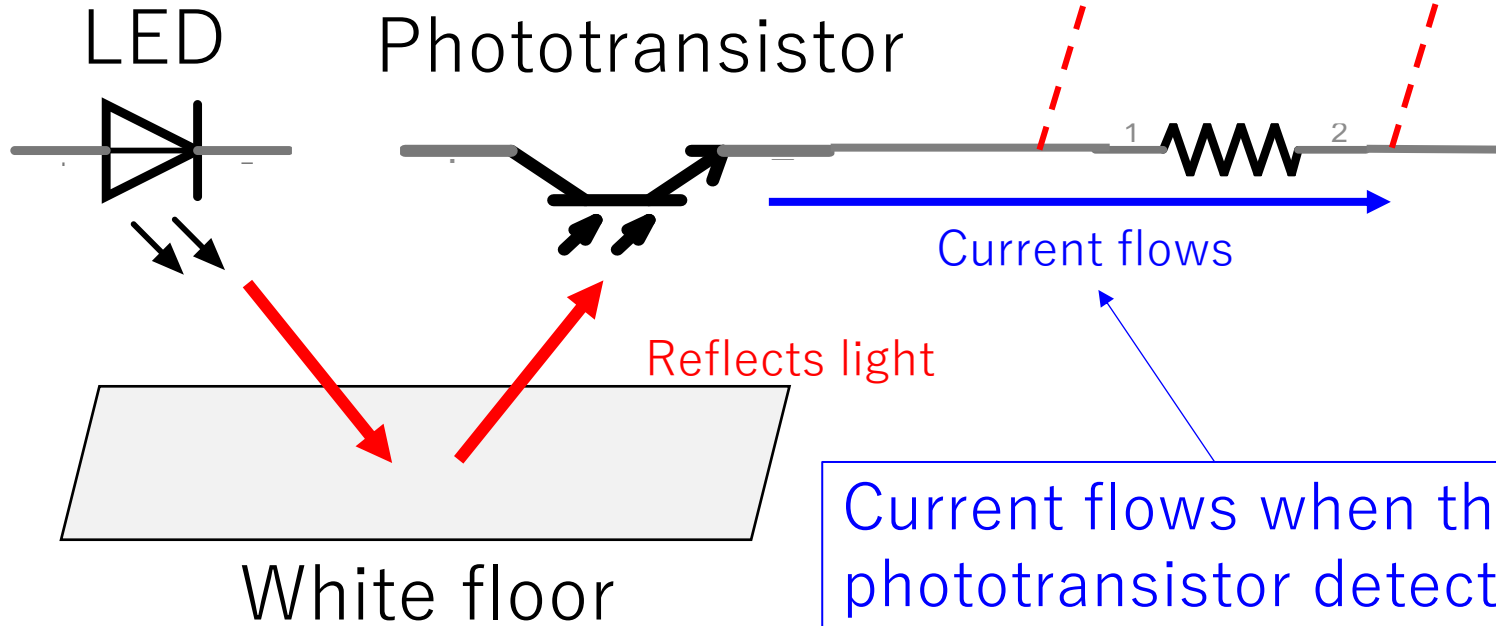
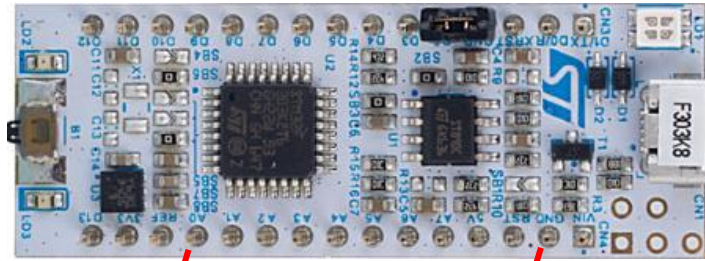
Current flows
when light is
detected

LBR-127HLD



Infrared : 赤外線

How to detect light

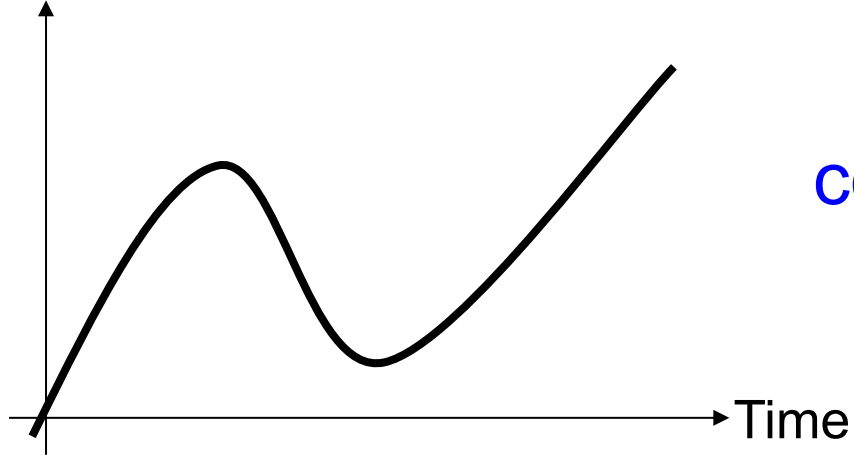


Measure voltage over a resistor to detect the current using a microcontroller

Current flows when the phototransistor detects light reflected on the white floor

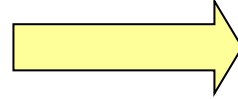
Analog-to-Digital converter (ADC)

Analog data



Any values (0-3.3V)

A/D
converter



Digital data

⋮
0000
⋮
0010
⋮
0100
⋮
0011
⋮
0001
⋮
0001
⋮
0010
⋮
0100
⋮
0101

Integers $0 \sim 2^{12} - 1 = 4095$
(in the case of 12bit ADC)



AnalogIn API

Return a **decimal number**
(0-1) by dividing 4095

Your microcontroller board (Nucleo-F303K8) changes a voltage (0-3.3V) to a 12 bit binary number. Then return a decimal number (0-1) if you use “AnalogIn” API.

Mbed API : “AnalogIn”

[Click here for documentation](#)

Read voltage from a pin

Initialization:

```
AnalogIn object_name(pin_name);
```

Class

Functions of AnalogIn :

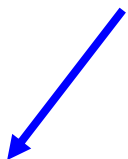
- read()
- read_voltage(),
etc.

Example:

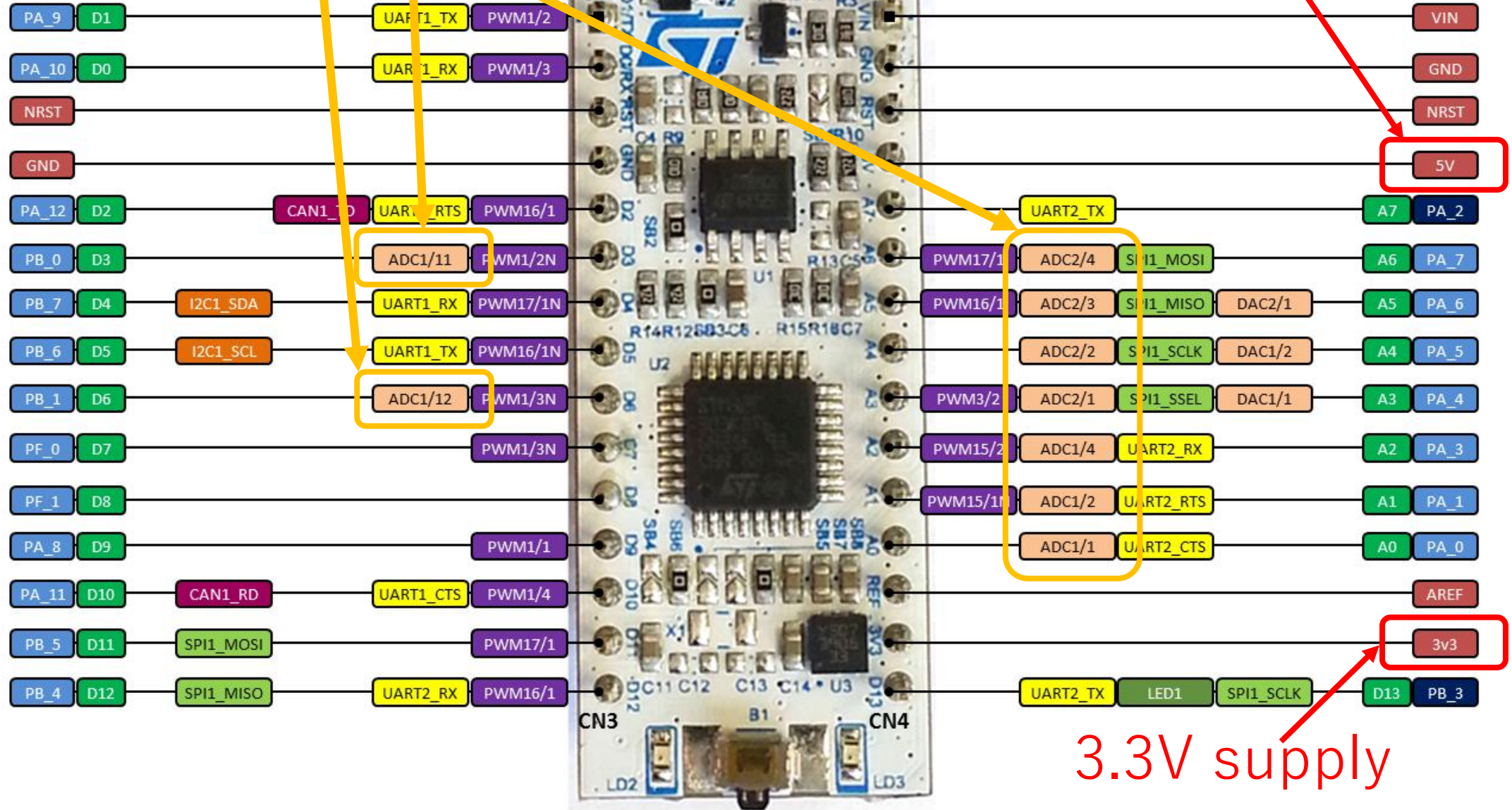
For input voltage 1.65V

object_name.read()  0.5

$0.5 \times 3.3 \text{ V}$



Pins for AnalogIN



Lead wires of photointerrupter

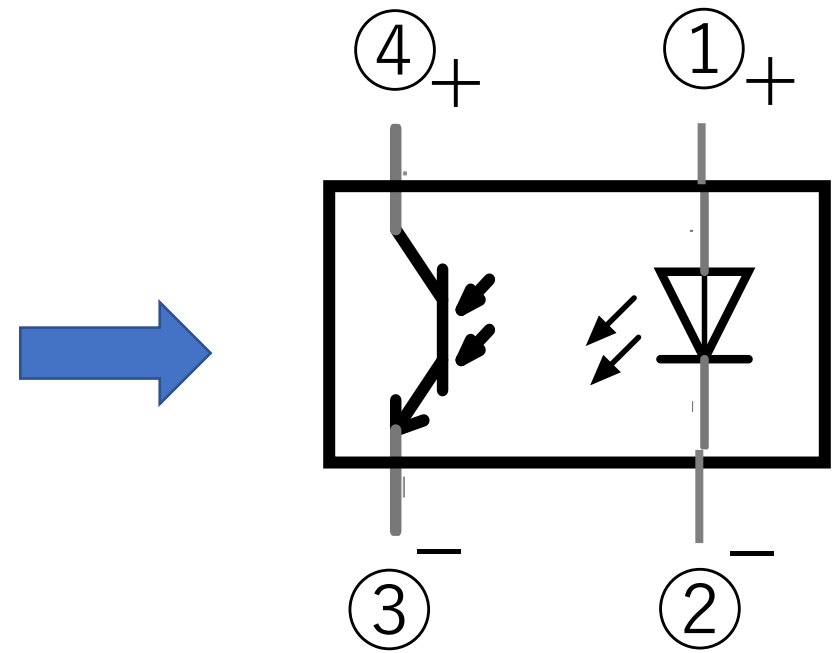
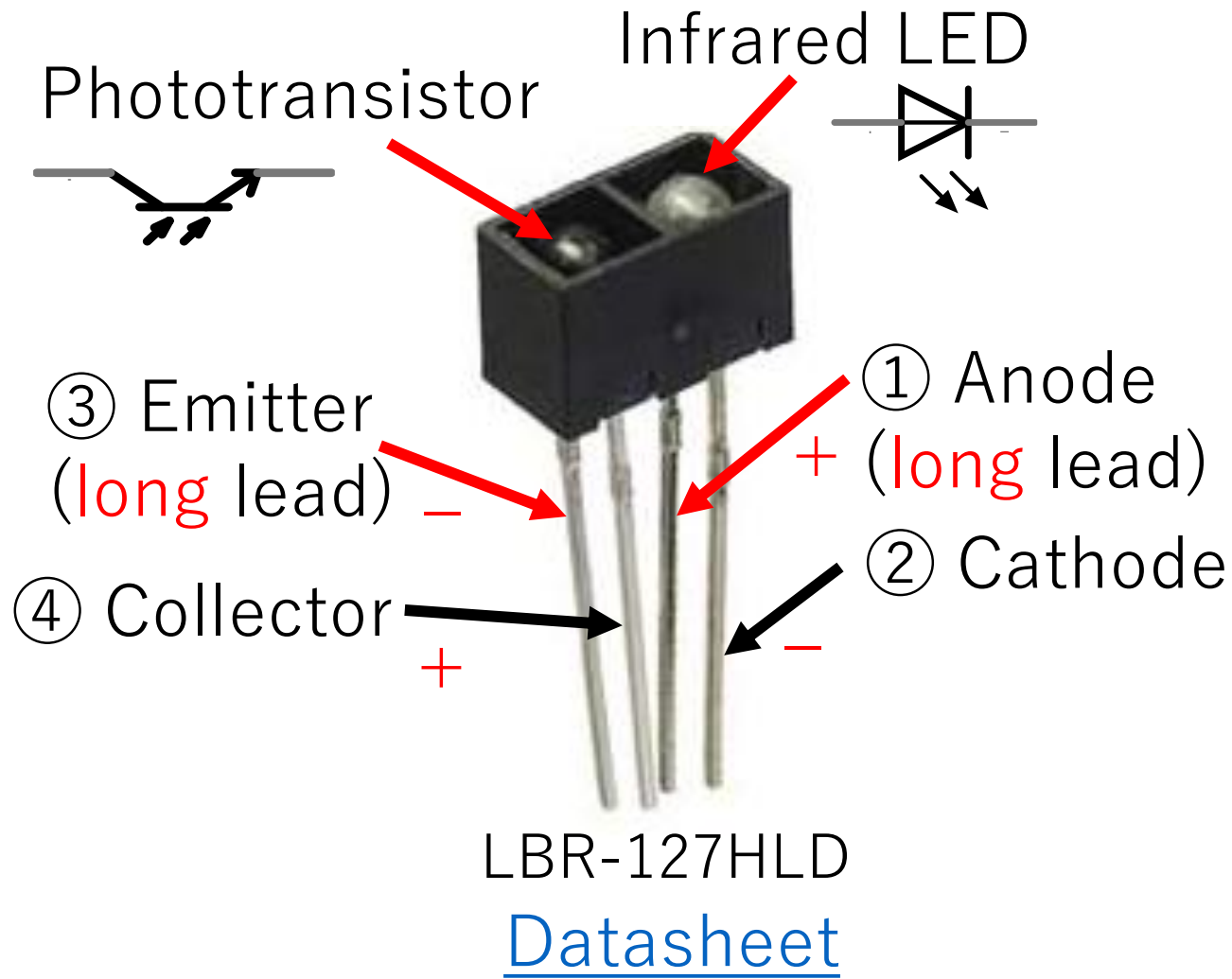


Diagram from datasheet

Be careful about the difference from the actual placement (① ~ ④)!!

Current limiting resistor for LED

Forward voltage of LED part (from [Datasheet](#))

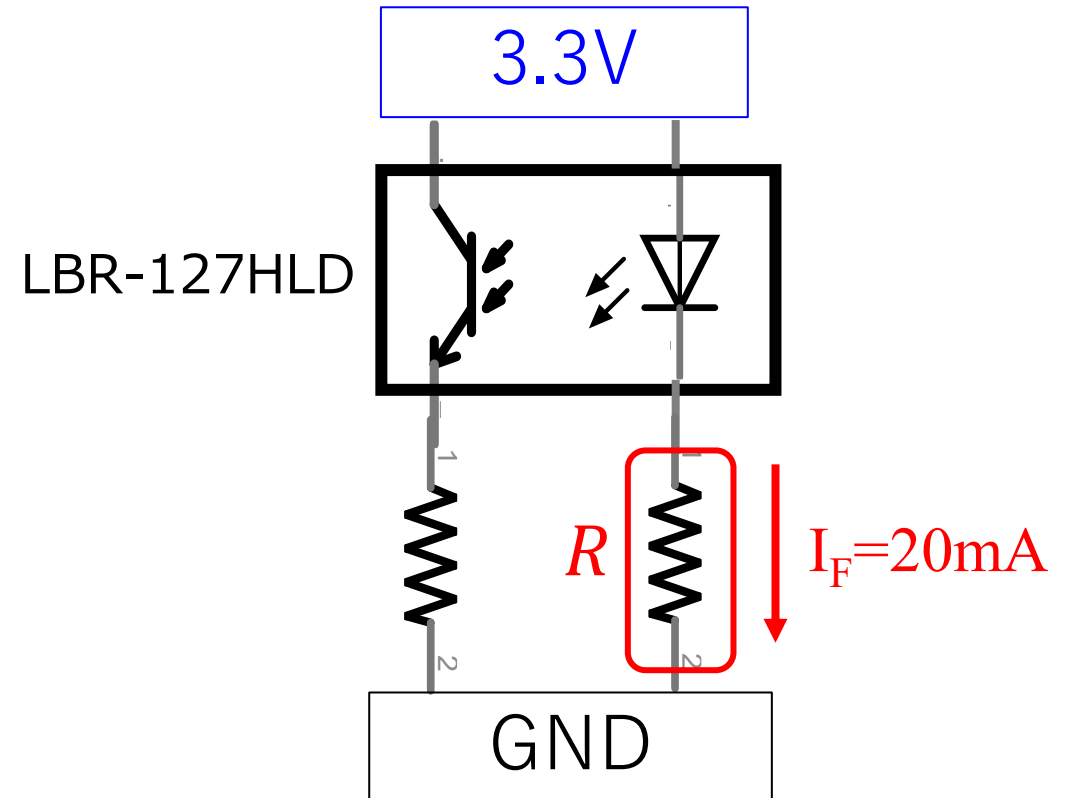
Item	Symbol	MIN.	TYP.	MAX.	Units	Conditions
Forward voltage	V_F		1.2	1.5	V	$I_F=20\text{mA}$

Maximum forward current: 60mA
(Ambient temperature: 25°C)

From Ohm's law,

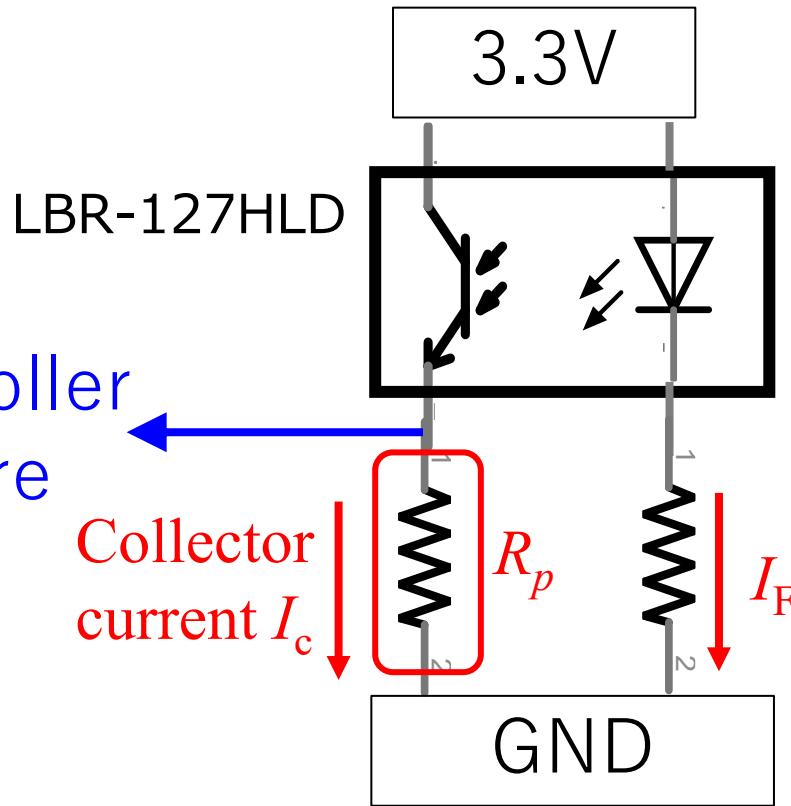
$$R = \frac{3.3 - 1.2}{20 \times 10^{-3}} = 105\Omega$$

↓
100 Ω
Available



Resister for phototransistor

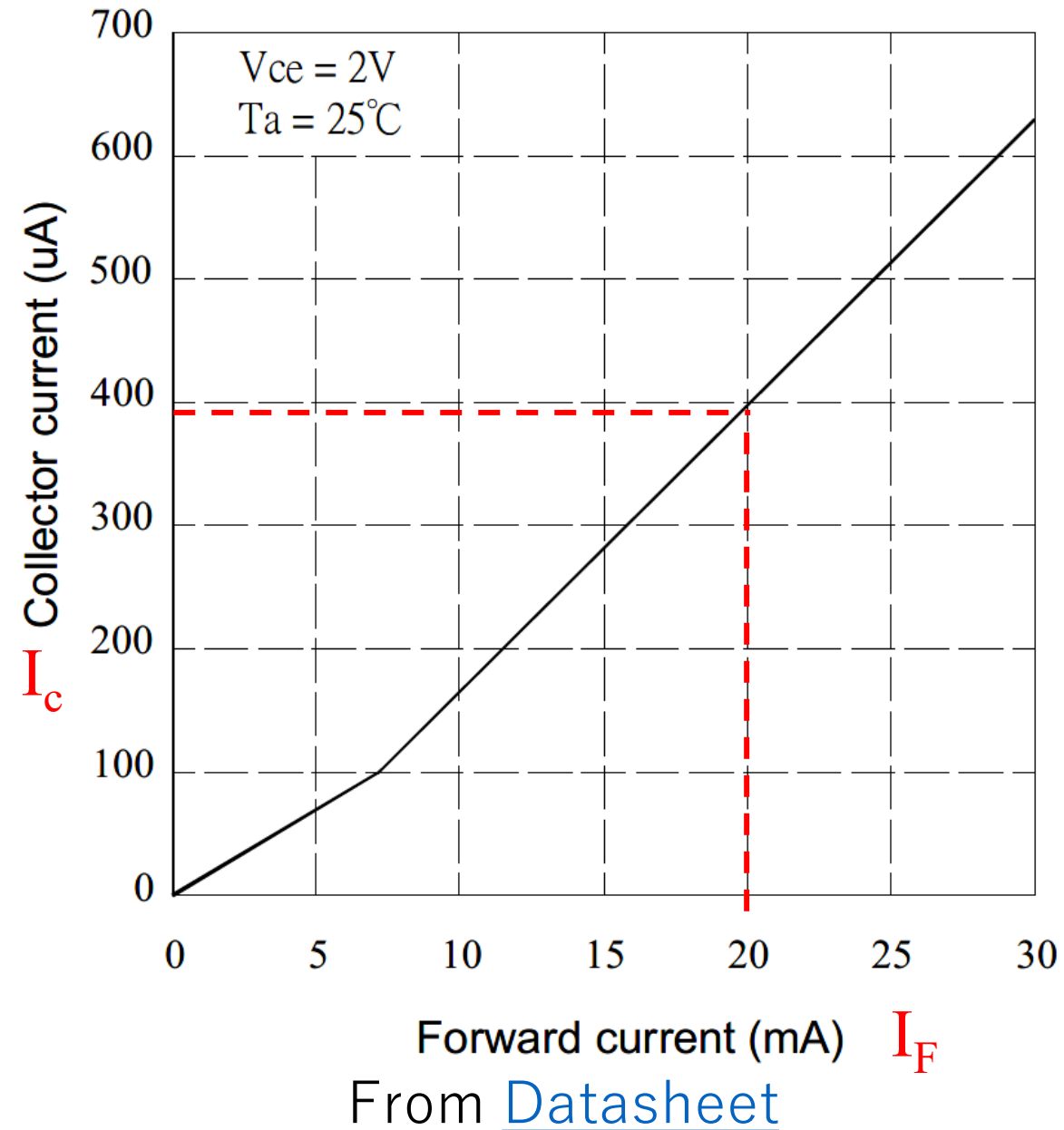
Microcontroller
can measure
 $0 \sim 3.3V$



For $I_F = 20\text{mA}$ ($I_c = 0.4\text{mA}$),

$$R_p = \frac{3.3[V]}{0.4[\text{mA}]} = 8.25 \text{ k}\Omega$$

8.2 k Ω



Simple example for “printf()”

```
6  #include "mbed.h"
7
8  int main()
9  {
10     float x = 7.0;
11     int i = 50;
12     printf("The float value is : %f \n", x);
13     printf("The float value is : %.2f \n", x);
14     printf("The integer value is : %d \n", i);
15 }
```

Today's exercises

Exercise 1-1: Blink LED on microcontroller board

Exercise 1-2: Blink LED on breadboard

Exercise 1-3: Display messages on serial monitor

Exercise 1-4: Variable resistor

Exercise 1-5: PWM signal

Exercise 1-6: Photointerrupter

Exercise 1-7: Photointerrupter & LED

Show the result of each exercise to an instructor or TA, every time you complete an exercise

Exercise 1-6

Make a circuit and a program that turns ON a LED, when a black line is in front of a reflective photo-interrupter

This is a combination of Exercises 1-2 and 1-6